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SPRING AI

GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

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Corso avanzato per Venis S.p.A, Venezia, Italia

Novembre 2025

MULTIMODALITÀ

SPEECH-TO-TEXT

- ➔ Processo di produzione testuale da audio del parlato
- ➔ Compatibilità Spring AI (al 11/2025)
 - ➔ OpenAI API
 - ➔ Azure OpenAI
 - ➔ OpenAI-compatible API (limitata)

Dipendenze applicativo

```
...  
<dependency>  
  <groupId>org.springframework.boot</groupId>  
  <artifactId>spring-boot-starter-web</artifactId>  
</dependency>  
<dependency>  
  <groupId>org.springframework.ai</groupId>  
  <artifactId>spring-ai-starter-model-openai</artifactId>  
</dependency>  
...
```

Configurazione applicativo

```
spring:  
  application:  
    name: demo  
  ai:  
    openai:  
      api-key: ${OPENAI_API_KEY}
```

Controller di trascrizione

```
@RestController
@RequestMapping("/api")
public class AudioController {

    private final OpenAiAudioTranscriptionModel openAiAudioTranscriptionModel;

    AudioController(OpenAiAudioTranscriptionModel transcriptionModel,
        SpeechModel speechModel) {
        this.openAiAudioTranscriptionModel = transcriptionModel;
        this.speechModel = speechModel;
    }

    @GetMapping("/transcribe")
    String transcribe(@Value("classpath:filename.mp3") Resource audioFile) {
        return openAiAudioTranscriptionModel.call(audioFile);
    }
}
```

- ➔ Servizio *speech-to-text* Gemini (≥ 2.5 -flash)
 - 1 Modifica configurazione di sistema
 - 2 Creazione interfaccia ed implementazione del servizio
 - 3 Modifica controllore MVC
 - 4 Test delle funzionalità con Postman/Insomnia

Configurazione applicativo

```
spring:
  application:
    name: demo
  ...
  ai:
    ...
    openai:
      ...
      audio:
        transcription:
          options:
            model: gemini-2.5-flash
```


Interfaccia servizio

```
package it.venis.ai.spring.demo.services;  
  
import org.springframework.core.io.Resource;  
  
import it.venis.ai.spring.demo.model.Answer;  
  
public interface MultiModalityService {  
  
    Answer getTranscriptionFromAudioFile(Resource audioFile);  
  
}
```

Implementazione servizio - I

```
package it.venis.ai.spring.demo.services;

...

@Service
public class MultiModalityServiceImpl implements MultiModalityService {

    private final ChatClient geminiChatClient;

    public MultiModalityServiceImpl(@Qualifier("geminiChatClient") ChatClient geminiChatClient) {

        this.geminiChatClient = geminiChatClient;
    }

    @Override
    public Answer getTranscriptionFromAudioFile(Resource audioFile) {

        String contentType = getContentType(audioFile);

        Media audioMedia = new Media(
            MimeTypeUtils.parseMimeType(contentType), audioFile
        );

        return new Answer(this.geminiChatClient.prompt()
            .user(u -> u.text("Trascrivi il seguente file audio: ").media(audioMedia))
            .call()
            .content()
        );
    }

    ...
}
```

Implementazione servizio - II

```
...  
  
private String getContentType(Resource audioFile) {  
    String contentType = URLConnection.guessContentTypeFromName(audioFile.getFilename());  
  
    if (contentType == null || contentType.equals("application/octet-stream")) {  
        String fileName = audioFile.getFilename();  
        if (fileName != null) {  
            String extension = fileName.substring(fileName.lastIndexOf('.') + 1).toLowerCase();  
            contentType = switch (extension) {  
                case "mp3" -> "audio/mpeg";  
                case "wav" -> "audio/wav";  
                case "m4a" -> "audio/mp4";  
                case "ogg" -> "audio/ogg";  
                case "flac" -> "audio/flac";  
                default -> "audio/mpeg";  
            };  
        } else {  
            contentType = "audio/mpeg";  
        }  
    }  
  
    return contentType;  
}
```

Implementazione controllore REST

```
package it.venis.ai.spring.demo.controllers;

...

@RestController
@Configuration
public class QuestionController {

    ...
    private final MultiModalityService multiModalityService;
    ...

    public QuestionController(QuestionService service,
                             RAGService ragService,
                             TimeToolsService timeToolsService,
                             MultiModalityService multiModalityService,
                             ...) {

        ...
        this.multiModalityService = multiModalityService;
        ...
    }

    ...

    @PostMapping("/gemini/ask/multi-modality/transcribe")
    public Answer getTranscriptionFromAudioFile(@Value("classpath:Venis_descrizione_azienza.wav") Resource audioFile) {
        return this.multiModalityService.getTranscriptionFromAudioFile(audioFile);
    }
}
```

<https://github.com/simonescannapieco/spring-ai-advanced-dgroove-venis-code.git>

Branch:

16-spring-ai-gemini-ollama-multimodality-audio-transcription